

One

THE BIRTH OF THE BIKE

It was nearly 10:30 p.m., and the artificial horse was nowhere in sight. A reporter had heard that the miraculous new machine was coming, though, so he waited expectantly in Philadelphia's Washington Square, among fashionable citizens out for a stroll after a long day of rain. It was May 1819, and the park—planted with trees, laid out with pathways, and surrounded by a white picket fence—was lit by candle-lantern streetlights and a misty half-moon.

Suddenly, the beast appeared, urged forward by a rider making huge, swinging strides with his legs. It went fast, probably about six miles an hour, a typical runner's pace. In the early nineteenth century, no one traveled that fast easily for long without the help of a horse.

The fascinated reporter took off after the rider, but before he could get far, someone in the park shouted, "Here comes another!" and a second rider flashed by him "like lightning," the reporter wrote. "Having a great desire to examine this curious machine, I exerted myself to overtake it, and luckily as I was almost exhausted the rider stopped, but the crowd collected around him, so I could only get a faint idea of its construction."

This amazing vehicle was a draisine, the most direct ancestor of the bicycle. Though American newspapers had been writing about the European invention for months, the first one had appeared in Philadelphia only the previous week. Roughly the same height as a modern bike, the draisine consisted of two in-line, wagon-spoked wheels connected by a curving horizontal bar that acted as a frame. There was a tiller attached to the front wheel for steering, and a saddle in the center, but there were no pedals. Instead, a rider straddled the saddle, gripped the tiller, and propelled the draisine like a scooter. With each step, the rider pushed off against the ground, allowing the draisine to coast until it needed a push with the other leg. Those who first saw the machine in action compared it to ice skating on the road. A typical draisine could weigh fifty pounds or more, twice the weight of an average bicycle today, so there wasn't much coasting uphill. Downhill or on level ground, though, it could move.

The draisine had been invented two years earlier in Germany by its namesake, Karl von Drais, a civil servant whose well-connected father had arranged for his release from the daily duties of his job as a forester. Times were tough in Europe: the 1815 eruption of Mount Tambora in what is now Indonesia had kicked up a persistent dust cover over the Northern Hemisphere that significantly cooled parts of the continent, causing freezes throughout 1816—the so-called year without a summer—and famine beyond that. In 1817, a feed shortage in Germany forced the slaughter of many horses. And it was in this context that Drais invented his man-powered vehicle, a sort of substitute horse. Though it could not haul loads or power machines, as horses could, the draisine could carry a rider.

Drais first demonstrated the wooden machine in Mannheim in the summer of 1817, and before long he was selling plans for the devices, and German mechanics were building and selling

knockoffs. In the following year, Drais patented the draisine in France, where it made a splash among the elite under the new name of “velocipede,” constructed from the words for “swift” and “foot.”

Crossing the channel to England, the draisine was revamped in 1818 by a London coach maker named Denis Johnson. With larger wheels and some iron parts in place of Drais’s wooden ones, the machine was renamed a “hobby-horse” or “dandy charger,” after the flamboyantly dressed young men who affected them.

In the United States, the draisine made its first public appearance in February 1819, when a musical-instrument maker in Maryland exhibited a specimen he had built based on a European drawing. The maker, James Stewart, advertised the device as “a new mode of travelling, combining the advantages of carriage, horse, and foot” and displayed it in Baltimore’s Concert Hall, charging twenty-five cents admission, the equivalent of about \$4.60 today. Originally open only during the day, the exhibit eventually added evening hours to accommodate the wives and children of men who had seen it.

One intrigued viewer was Philadelphia’s Charles Willson Peale, a nationally famed artist and naturalist. The seventy-seven-year-old Peale, best known for painting portraits of George Washington and other Revolutionary War heroes, was also an avid amateur zoologist and paleontologist. Peale founded a private museum of science and curiosities that showcased a mastodon skeleton he had personally extracted piece by piece from a sloppy bog on a New York farm. (At its unveiling, the skeleton display was only the second fossil reconstruction in the world.)

In early 1819, Peale came out of artistic retirement to visit Washington, DC, and paint President James Monroe and other

dignitaries. On the way home afterward, he stayed for several days in Baltimore, where he encountered the amazing mechanical horse. Peale wanted one.

After returning to his estate on the outskirts of Philadelphia, he found a British illustration of the draisine in a local shop and based the design of his own on it, hiring a blacksmith to make the machine from scrap iron taken from an old grain thresher. Peale deposited the machine in his museum, where it earned “a very considerable profit” in admission fees and inspired imitators, as he wrote to his son Rembrandt. “As soon as it was heard of, several of them appeared immediately, constructed in different manners of wood,” he added.

Peale’s iron draisine was one of the two “curious machines” that so fascinated the reporter that May night in Washington Square. And the riders were probably Peale’s sons Rubens and Franklin, according to a letter Peale wrote a friend. The young men had likely liberated the iron draisine from its display after hours to show it off, along with a wooden one built by Franklin.

The draisine wasn’t just a curiosity, though. It was a kind of jet pack in an immobile age. Even Peale’s clunky metal specimen—which amounted to deadweight when pushed up a hill—flew downhill “like the very devil,” as yet another son, Charles Linnaeus, put it to his baby brother Titian. (Among his other activities, Charles Willson Peale fathered seventeen children, naming most of the boys after artists.) Peale himself was too afraid of falling to coast down hills, but that summer he wrote of watching younger men on Franklin’s wooden draisine tear down them “in a swiftness that dazzles the sight.” The machines bumbled along fine on flat land, too, allowing riders to scoot and glide as fast as nine miles per hour, a horse’s trot.

Only the well-to-do had the resources and leisure time to build or buy such machines, just as only the well-to-do could afford to keep horses. But the sight of draisine riders inspired

predictions of a form of personal mobility that would be affordable to all. A newspaper reporter who had heard of a velocipede ride from Germantown—the site of Peale’s estate, Belfield—to Philadelphia during the summer of 1819 wrote that “several grave and learned gentlemen” rode the eight miles “in as short a time as real horses could have done.” That year saw a financial panic in the United States, and horse fodder was expensive, the reporter noted. “We should be happy to see these ‘animals’ introduced into this country. . . . Whilst oats are selling at sixty-two cents a bushel, it would be a serious saving to our Sunday equestrians.”

During the rest of the year, a series of American cities picked up the velocipede fashion. Boston, New Haven, New York, Washington, DC, and Savannah saw the machines exhibited or rented by the day or month in public parks. Specimens were spotted as far west as Cincinnati, then a frontier boomtown. Newspapers were full of mentions. The sensation of balancing in a line by shifting body weight was new to most Americans—ice-skating being a seasonal pursuit for Northerners willing to risk breaking through the ice and drowning—and riding schools cropped up to serve the need. “A few lessons are sufficient to overcome the difficulties necessarily attendant on its novelty,” wrote a New York newspaper contributor identified as *Gymnasticus*, “after which it is not only agreeable, but even fascinating.”

But for all its fascination, the draisine died a quick death, both in the United States and in Europe. It is estimated that fewer than a thousand were ever made in this country, which had a population of more than nine million, and their use was restricted almost from the start after an outcry over collisions with pedestrians. In July 1819, for example, a New York man wrote to a newspaper about a four-year-old boy he had seen knocked down and injured by two fops riding velocipedes on the sidewalk. “Is it not enough to be annoyed by mud, dust,

chimney sweeps, hogs, mad dogs, hot corn [that is, corn on the cob hawked by street urchins], and all the evils necessarily attendant on a great city like ours," he asked, "but that our peaceable citizens, our wives and our children, cannot enjoy a walk in the evening, without the danger of being run over by some of these new-created animals?" Within months, draisine riding was against the law on most American city sidewalks.

With sidewalks off-limits, you might think that riders would simply take to the roads, but in most places in America, that wasn't a realistic option. Most city streets weren't paved, and when they were, it was usually with cobblestones or bricks, which made for a bumpy ride. Not to mention the obstacle course created by horses and their droppings. Unpaved city streets were rutted and dusty in dry weather, gloppy after a rain.

In the countryside, it was even worse. More than 70 percent of the US population still lived on farms, and rural roads, where they existed, were notoriously bad, "hardly more than broad paths through the forest," according to a 1951 economic history. "In wet places, they presented a line of ruts with frequent mud holes, and, where dry, a powdered surface of deep dust." One early Ohio law decreed that stumps left in the road should be no more than one foot high. Coaches and wagons routinely got stuck in ditches and overturned or broke their axles, which made overland travel slow going; in 1813, a cargo wagon drawn by three horses was noted for taking only two weeks to make the trip from Boston to Philadelphia, "notwithstanding the wretched state of the roads," as a Baltimore business journal reported. Unless you had a hilly country estate to get around, as Peale did, you couldn't count on draisines for transportation. And if you had an estate, you also had horses to ride.

Then too, the heavy velocipede was a workout. The British cycle historian Cally Callomon has called it "the pogo stick of its day," more suited for recreation than transportation. As a

horse substitute, therefore, it couldn't last. Within a year of its introduction in the United States, it had nearly disappeared, as it soon did in Britain. In May 1820, a wag rewrote Hamlet's soliloquy as

To ride—or not to ride—that is the question—
Whether 'tis nobler in the mind to flatter
The whims and follies of outrageous fashion—
Or to ride horseback, and not stride away
On a Velocipede!

He published the poem in a Boston paper with the caveat that the velocipede was “not yet wholly obliterated from the minds of all,” an admission that it had been forgotten by most. And even that poem, in the end, came down on the side of not riding the mechanical horse.

The word “velocipede” stayed in use during the following decades, referring to a series of three- or four-wheeled carriages that inventors fitted out with treadle or hand-lever drives, none particularly efficient. But the two-wheeled, human-powered vehicle was widely agreed to be a technological dead end.

Meanwhile, Americans continued to develop other ways of getting around. Some private highways were built starting in the 1790s; these were called “turnpikes” because of the pointed spikes that gatekeepers turned aside for travelers after their tolls were paid. And the federal government built a road from Maryland to Illinois to encourage westward development. But the quality of these roads varied. There was no such thing as asphalt pavement; some early turnpikes were built on solid stone, and others consisted of gravel or just dirt. In any event, turnpikes connected only certain cities. Elsewhere, rural roads tended to serve a few communities and then just stop, lending truth to what would later be a punch line: “You can't get there from here.”

In America, the chief method of long-distance travel in the early nineteenth century was by water. Most trips between East Coast cities were from seaport to seaport, and in the continent's interior, rivers and lakes were the only paths to take; there were no connecting roads at all. (This is why Americans came to use the verb "to ship" for "to send a package.") Originally, traveling upstream meant rowing, sailing, or using poles as leverage against the river bottom, but steamboats—first invented in 1787—gained power and popularity early in the nineteenth century. By 1820, there were fifty steamboats operating on the Mississippi River, churning upstream at the unheard-of rate of ten miles an hour. That year, one small steamboat that began running regular trips on the Ohio River was named the *Velocipede*.

The river voyage from New Orleans to Cincinnati, which took ninety days before the use of steam power, "is now performed, with the greatest ease and safety, in eleven or twelve days," wrote one Englishman who made the trip in 1827. ("Safety" was a relative term: steamboat boilers routinely exploded, killing passengers and crew alike.) The speed of steamboat travel fit the restless American spirit, one French visitor wrote in 1835. The typical American "is devoured with a passion for locomotion, he cannot stay in one place," the visitor opined. "He is always disposed to emigrate, always ready to start in the first steamer that comes along."

Boats were getting faster, and water routes were multiplying. At around the same time as the draisine's brief appearance, America embarked on a frenzy of canal construction. Starting with the Erie Canal's groundbreaking in 1817, states and private investors began digging channels linking Atlantic Ocean tide-water areas to the Ohio and Mississippi Rivers and the Great Lakes. These new water avenues eased inland travel and vastly reduced the cost of bringing goods to market. Too narrow and shallow for early steamboats, canals were traversed by barges

hitched to horses or mules that trudged along towpaths on the banks. In 1827, our English traveler continued from Cincinnati by stagecoach to Lake Erie, and from there by boat to New York City via the partially completed Erie Canal, which he described as “crowded with boats of considerable size, laden with the various produce of the western and northern states, and returning with numerous emigrants, moving westward with their families and effects.” Starting with only about 100 miles of canals in 1816, the nation had developed more than 3,300 miles’ worth by 1840.

And of course railroads entered the picture in 1830, when the Baltimore and Ohio opened its first thirteen miles of track. Built mainly by private investors, with liberal assistance from states, these transit lines developed in a scattering of short segments that, over time, began to connect and thereby threaten the primacy of water transit. Though rickety by modern standards, trains were often twice as fast as steamers, and they could operate year-round; boats were stymied by frozen winters in the North and parched summers in the West. By 1840, there were almost exactly as many miles of railroad track in the United States as there were of canals: more than 3,300. But while domestic canal construction never went much further—in part because most of the obvious connections between waterways had already been made—railroad builders kept laying track.

Even more than canals and steamboats, trains seemed to collapse the country’s vast distances by shrinking the time it took to cross them. In 1859, a Boston writer predicted that the Adams Express Company would soon be able to send parcels from New Orleans to New York via rail in four days: “How much such extraordinary dispatch by the Adams Express will do to increase the trade and communication between the two great cities of the North and Southwest, it is impossible, of course, to estimate; but we look for wonders.”

Just two years later, the Civil War severed trade and communication between North and South for four gruesome years and left many Southern cities and rail lines in ruins. But the perception that technology—including the telegraph, introduced in 1844—was shrinking distances and accelerating American life endured. People called the phenomenon “the annihilation of space and time.”

In July 1865, a few months after the South surrendered at Appomattox, a French mechanic by the name of Pierre Lallement arrived in Brooklyn, bearing the parts of a machine that he had built in Paris. The seventy-pound device, which had wooden wheels and a wrought-iron spine, was very similar to an old-timey two-wheeled velocipede he had seen in France, but with one key addition: there were foot pedals attached to its front wheel. Lallement, a quiet, dark-haired, unbrowed young man, soon moved to Ansonia, Connecticut, for a factory job, and he continued tinkering with the device until it was ready for short rides to and from his workplace.

That fall, Lallement conducted a road test of about four and a half miles, pedaling the velocipede mostly uphill to the nearby village of Birmingham (now part of Derby) and then doubling back home to Ansonia. As he told a journalist twenty years later, his delight during one bumpy downhill stretch turned to panic when he realized that his brakeless vehicle was about to rear-end a horse-drawn wagon. He yelled a warning to the two men in the wagon, then veered and tumbled into a roadside culvert filled with water, cracking his head in the process. The terrified men, meanwhile, whipped their horses into a run and took off.

Lallement collected himself and rode to Ansonia’s main street, where, drenched and bleeding, he stopped in a tavern. “There he found the two men,” the journalist wrote, “relating between drinks how they had seen the dark Devil, with human head and body half like a snake, and half like a bird, just hover-

ing above the ground which he seemed no way to touch, chase them down the hill.” Lallement approached the men and, in his thick French accent, exclaimed, “I was the Devil!”

Lallement secured a US patent for the device in 1866, but wasn’t able to attract a manufacturer, and he returned home to France in early 1868, leaving his velocipede behind. When he arrived in Paris, though, he discovered that others were making and selling a pedal-cranked velocipede very similar to the one he had left in the States. Worse yet, one of the manufacturers was claiming to have invented it. How this happened is still “murky,” says David Herlihy, author of the indispensable 2004 book *Bicycle: The History*. Herlihy thinks it likely that one of the French maker’s backers had seen Lallement riding his pedaled velocipede in Paris years before and decided to copy it.

Whatever happened, the French claimed ownership of the crank-propelled velocipede—by then also called a “bicycle,” from the Latin word for “two” and the Greek word for “circle”—and a new craze erupted, both in Europe and in the United States. In the summer of 1868, the world-famous Hanlon Brothers acrobatic troupe included velocipede races in their act during a North American tour. Soon, American carriage makers, wheelwrights, and blacksmiths were producing them.

Americans greeted the pedal-cranked velocipede with the same shocked amazement as the two Connecticut men who had witnessed Lallement’s diabolical downhill ride. Spectators filled Cincinnati’s Pike’s Music Hall for an exhibition race in January 1869, “all agog to fathom the mysteries of the wondrous bicycle,” a local paper reported. Few people alive had seen the velocipede’s predecessor of fifty years prior, the draisine, in action—and those who had seen it could understand how that old machine operated. This was different, though. A rider on a pedal-cranked velocipede could seemingly go for as long as he wanted without visible means of support. And going faster

didn't make the machine wobblier, as one might expect; paradoxically, it made it steadier. As a reader wrote in a letter to *Scientific American* in 1868, "That a carriage or velocipede with but two wheels . . . should maintain an upright position is, to the superficial observer, one of the most surprising feats of practical mechanics."

Cycling rinks and academies cropped up across the country as the public clamored to learn how to ride. *Scientific American* diagnosed a "velocipede mania" and proclaimed that "walking is on its last legs." This, finally, was a machine for shrinking the city as earlier machines had shrunk the countryside.

"Space has been a sort of enemy to enterprise, and a great part of our energies has been devoted to what is called the annihilation of space," wrote the editors of a bicycle-industry magazine in early 1869. "To this end steamboats have been invented, railroads have been projected, which answer very well for the longer distances, where many miles have to be traversed, but for the shorter distances, up to the invention of the velocipede, we have had nothing adequate to our wants." Stagecoaches were crowded with germey strangers and didn't take you door-to-door, the editors added, and horses were expensive and couldn't be risked on slippery pavement. "The two-wheeled velocipede is the animal which is to supersede everything else."